



Neural network and support vector machine for the prediction of chronic kidney disease: A comparative study

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ABSTRACT

This paper aims to assist in the prevention of Chronic Kidney Disease (CKD) by utilizing machine learning techniques to diagnose CKD at an early stage. Kidney diseases are disorders that disrupt the normal function of the kidney. As the percentage of patients affected by CKD is significantly increasing, effective prediction procedures should be considered. In this paper, we focus on applying different machine learning classification algorithms to a dataset of 400 patients and 24 attributes related to diagnosis of chronic kidney disease. The classification techniques used in this study include Artificial Neural Network (ANN) and Support Vector Machine (SVM). To perform experiments, all missing values in the dataset were replaced by the mean of the corresponding attributes. Then, the optimized parameters for the Artificial Neural Network (ANN) and Support Vector Machine (SVM) techniques were determined by tuning the parameters and performing several experiments. The final models of the two proposed techniques were developed using the best-obtained parameters and features. The empirical results from the experiments indicated that ANN performed better than SVM, with accuracies of 99.75% and 97.75%, respectively, indicating that the outcome of this study is very promising.

1. Introduction

Kidney diseases are disorders that affect the functions of the kidney. During the late stages, kidney diseases can cause kidney failure. According to the data most recently released by the Saudi Center of Organ Transplantation Registry, 10,203 patients diagnosed with kidney disease receive hemodialysis [1]. Diabetes, high blood pressure, and unhealthy lifestyles have led to an increase in the number of patients with CKD [2]. Patients with CKD suffer from various side effects. These complications include damage to the nervous and immune systems that disrupt daily activities. To assist in the prevention of CKD, machine learning techniques can be utilized to diagnose CKD at an early stage.

Artificial Neural Network (ANN) and Support Vector Machine (SVM) are widely used machine learning techniques. Both ANN and SVM have advantages and have been proven to perform excellently in several fields, including medical diagnosis, weather prediction, stock market analysis, and image processing [3]. ANN is a mathematical model that acts similar to human neurons. ANN has the following two major features: the ability to learn how to perform its functions once it

is properly trained and the ability to generalize and produce a reasonable solution to unobserved data [4]. In contrast, SVM is a computational algorithm that can learn from experience and examples to allocate labels to objects. The basic function of SVM is to separate binary labeled data based on a line achieving the maximum distance between the labeled data [5]. SVM performs well and has good accuracy even with limited examples. Due to its fruitful advantages, SVM is distinct from other machine learning techniques [6–8]. This paper applies both ANN and SVM techniques to a CKD dataset to measure their accuracy and compare their performance. The WEKA was used to help in pre-processing and conducting the experiment using the dataset. The experimental results indicated that ANN outperformed SVM in almost all models of this dataset. ANN showed a better performance with an accuracy of 99.75% using the optimized features, while SVM had a performance accuracy of 97.75%. The ANN classifier had a considerably longer runtime than SVM, but no large difference in accuracy was observed.

The remainder of this work is organized as follows. Section 2 provides a review of the related literature. Section 3 discusses the proposed

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